STUDY OF INFORMATION FLOW AMONG ECG LEADS SIGNALS

***Abstract*—The main scope of this research is to analyze the flow of information of ECG signals of 12 leads. This study examines the information provided by each electrocardiography (ECG) lead, in order to reveal which of them offers greater information to contribute to the diagnosis and treatment of cardiovascular diseases. The results highlight the importance of identifying the most informative leads, which could improve accuracy in detecting cardiac conditions, guide more informed clinical decisions and contribute to the design of future diagnostic devices. In order to achieve this, two mathematical tools such as mutual information and transfer entropy were applied. For the study, two independent databases were employed to made all the calculations and obtain repeatability of patterns as a kind of confirmation of the results. The conceptual simplicity of the used tools makes this proposal simple to implement, computationally efficient and practically straightforward to interpret.**

***Index Terms*—ECG, mutual information, transfer entropy, signal information flow**

I. Introduction

As Bayés de Luna says, "The electrocardiogram (ECG), introduced into clinical practice more than 100 years ago by Einthoven, constitutes a lineal recording of the heart's electrical activity that occurs successively over time" [1]. In this context, the utility of ECGs has been demonstrated not only for detecting problems and making diagnoses, but also for evaluating the progression of treatment and how the patient responds to it. It is important to note that a normal ECG does not imply that the patient is healthy and does not suffer from heart disease. Following [1], in ECG recordings, usually six frontal (I, II, III, VR, VL, VF) and six horizontal (V1-V6) leads are used, and therefore the study and analysis of the information flow between the 12 leads will be crucial to better understand the problem and improve the accuracy of diagnosis, treatment, and follow-up. Although the information provided by the different leads of an ECG signal is not the same (see for example [1], [2]), given that they electrically reflect the hydro-mechanical behaviour of the cardiovascular system [3], [4], there may exist links among the different leads that make their use redundant when processing the compiled information of a given study. Therefore, the objective of this research was to analyze this information flow between leads using tools such as mutual information, which helps to determine the dependency between signals, that is, how correlated the signals from one lead are respect to other, and transfer entropy which is the amount of information that one random variable transfers to another over time, that is, the causal relationship between signals to understand how one signal affects or predicts another. To carry out this research, two independent databases were used; one of them called **PTB-XL, a large publicly available electrocardiography dataset**, a once restricted database which was publicly released in 2019, and another one used in an international challenge in ECGs classification named **The China Physiological Signal Challenge 2018: Automatic identification of the rhythm/morphology abnormalities in 12-lead ECGs**. These two databases proved useful to compare the results and to detect the most significant leads to provide insights in the selection of the electrodes location for the diagnosis and treatment of cardiovascular diseases. There are prior cases of leads selection for ECG, as for example reported in the work of Sameni et al. [5] , however the use of information flow tools for this purpose was not found in the available literature.

The authors wish to express their gratitude to the scholarship program for undergraduate and graduate students of the XXX University, and to the Secretariat of Science, Technology and Productive Innovation of the XXXX Faculty, and to the YYYYYY Research and Development Project YYYYYY.

II. Materials and Methods

Although there are other mathematical tools and other methodologies for the study of causality between signals (for more information see [6]–[8]) mutual information and transfer entropy have been used in this research, due to their conceptual clarity and efficiency in both computational implementation and computational performance. The mutual information was applied as a kind of first approach to detect links among the signals provided by the ECSs leads. Then the transfer entropy was used to determine the direction of the flux of information among the ECGs leads. In the following subsections there is a brief introduction to both concepts.

A. Introduction to Mutual Information and its significance

Mutual information is a fundamental concept in information theory that quantifies the statistical dependence between random variables [9], [10]. It provides a measure of how much information one random variable contains about another random variable, capturing the degree of correlation or association between them. In the context of statistical dependence, two random variables are considered to be dependent if knowing the value of one variable provides information about the value of the other variable. Mutual information precisely quantifies this dependency by measuring the reduction in uncertainty about one variable due to the knowledge of the other variable. By quantifying the statistical dependence between random variables, mutual information provides valuable insights into the underlying structure and dynamics of complex systems [11]. Its ability to capture the information shared between

variables makes it a fundamental measure in various scientific disciplines, driving advancements in research and practical applications [12].

Mutual information, denoted as $I(X,Y)$, measures the amount of information that one random variable, X, contains about another random variable, Y. Mathematically, it is defined as:

$$I(X,Y) = \sum_{x\in X}\sum_{y\in Y} p(x,y)\log\left(\frac{p(x,y)}{p(x)p(y)}\right), \quad (1)$$

where $p(x,y)$ is the joint probability distribution of X and Y and $p(x)$ and $p(y)$ are the marginal probability distributions. As shown in [2] the following calculations were made to estimate the dependence of the ECG signals according to the lead used to obtain the data.

$$I(X,Y) = \frac{1}{N}(\sum_{i=1}^{N}(\psi(k)+\psi(N)-\psi(n_x(i)+1)) - \sum_{i=1}^{N}(\psi(n_y(i)+1))), \quad (2)$$

where:

- $\frac{1}{N}\sum_{i=1}^{N}$ denotes the average over all N data points in the sample.
- $\psi(\cdot)$ is the digamma function, which is the logarithmic derivative of the gamma function.
- $\psi(k)$ is the digamma function evaluated at k, where k is the number of nearest neighbors used in the estimation.
- $\psi(N)$ is the digamma function evaluated at the sample size N.
- $\psi(n_x(i)+1)$ is the digamma function evaluated at $n_x(i)+1$, where $n_x(i)$ is the number of points within the distance to the k-th nearest neighbor in the marginal space of X for the i-th data point.
- $\psi(n_y(i)+1)$ is the digamma function evaluated at $n_y(i)+1$, where $n_y(i)$ is the number of points within the distance to the k-th nearest neighbor in the marginal space of Y for the i-th data point.

B. Introduction to Transfer Entropy

"As an asymmetrical measure, TE quantifies the amount of directed (time-asymmetric) transfer of information between random processes and therefore is related to the measures of causality" [14]. "Transfer entropy is capable of detecting directional coupling changes in non-linear biomedical time series analysis featuring a small number of observations and presence of outliers." [15]

Following Joon Lee et al. [15] given two concurrently sampled time series $X = \{x_1, x_2, ..., x_N\}$ and $Y = \{y_1, y_2, ..., y_N\}$, the transfer entropy from X to Y, denoted by $T_{X\to Y}$, can be derived from conditional entropies as follows:

$$T_{X\to Y} = \sum_{y_i, y_{i-t}^{(l)}, x_{i-\tau}^{(k)}} p(y_i, y_{i-t}^{(l)}, x_{i-\tau}^{(k)}) log\frac{p(y_i|y_{i-t}^{(l)}, x_{i-\tau}^{(k)})}{p(y_i|y_{i-t}^{(l)})}, \quad (3)$$

where i indicates a given point in time, τ and t are the time lags in X and Y, respectively, and k and l are the block lengths of past values in X and Y, respectively. In biomedical experiments it is useful to choose $k = l = t = 1$, simplifying the equation (3) to:

$$T_{X\to Y}(\tau) = \sum_{y_i, y_{i-1}, x_{i-\tau}} p(y_i, y_{i-1}, x_{i-\tau}) log\frac{p(y_i, y_{i-1}, x_{i-\tau})p(y_{i-1})}{p(y_{i-1}, x_{i-\tau})p(y_i, y_{i-1})}. \quad (4)$$

The main challenge in transfer entropy computing is calculating the joint probabilities in (4). The chosen methodology in this study was to use adaptive partitioning following the Darbellay-Vajda Algorithm combined with ordinal sampling (ranking). In ordinal sampling, the time series values in X and Y are substituted with their ranks in sorted X and Y. Let the transformed time series of X and Y be $U = \{u1, u2, ..., uN\}$ and $V = \{v1, v2, ..., vN\}$, respectively (U and V contain integers that represent ranks). Rewriting (4) in terms of U and V:

$$T_{U\to V}(\tau) = \sum_{v_i, v_{i-1}, u_{i-\tau}} p(v_i, v_{i-1}, u_{i-\tau}) log\frac{p(v_i, v_{i-1}, u_{i-\tau})p(v_{i-1})}{p(v_{i-1}, u_{i-\tau})p(v_i, v_{i-1})}. \quad (5)$$

After ordinal sampling, the D-V algorithm recursively partitions the three-dimensional space defined by v_i, v_{i-1}, and $u_{i-\tau}$ into cubes of varying sizes. The result of the partitioning process is a finite number of cubes, L, each of which contains nonzero data points. By approximating each probability in (5) by counting, $T_{U\to V}(\tau)$ can now be estimated using the L partitions as follows:

$$T_{U\to V}(\tau) \approx \sum_{k=1}^{L}\frac{n_k}{P} log\frac{n_k n_k^{v_{i-1}}}{n_k^{v_{i-1},u_{i-\tau}} n_k^{v_i,v_{i-1}}}, \quad (6)$$

where P is the total number of data triplets (v_i, v_{i-1}, and $u_{i-\tau}$), n_k is the number of data points in the kth partition, and $n_k^{v_{i-1}}$, $n_k^{v_i,v_{i-1}}$, and $n_k^{v_{i-1},u_{i-\tau}}$ are the numbers of data points in the entire data that are greater than or equal to lower bounds and less than upper bounds of the kth partition with respect to the dimensions in the superscript.

C. Databases

The first database used in this study was taken from "The China Physiological Signal Challenge 2018: Automatic identification of the rhythm/morphology abnormalities in 12-lead ECGs" [16]. This database is composed of 6877 12-leads

ECG recordings collected from 11 hospitals. The recordings last between 6 and 60 seconds and were sampled at 500 Hz. For this study, only registers classified as Normal (906 recordings) were used, they were cut down to 10 seconds and downsampled to 100 Hz. All recordings lasting less than 10 seconds were discarded.

The second database is "PTB-XL, a large publicly available electrocardiography dataset" [17], taken from the PhysioNet website [18], [19]. "The PTB-XL ECG dataset is a large dataset of 21801 clinical 12-lead ECGs from 18869 patients of 10 second length". These records were sampled at 100 Hz and 500 Hz. Given the fact that there is no need to have a high sampling rate such as 500 Hz, the 100 Hz records were used in order to reduce computing time. Only records classified as Normal (414 records) were used.

III. Development of the Proposal

In this study the following steps were pursued to compare the dependencies between the 12 leads of an ECG signals databases using mutual information estimation. The mutual information of each pair of leads was calculated using the estimator given by (2) and finally, to analyze the results, a mutual information matrix was created to visualize the dependencies and to interpret the dependencies between leads. If the results show a high mutual information values between leads this indicates strong dependencies which might imply redundant information or a physiological coupling. On the other hand a low mutual information indicates weak dependencies suggesting that the leads are more independent. By analyzing the mutual information matrix is possible to identify pairs of ECG leads with significant dependencies, which can provide insights into the underlying physiological processes or help in reducing dimensionality for further analysis by removing or combining redundant leads that provide similar information The analysis of the information flow between the different ECG leads is the key to select ECG leads to use in several applications, such as the possible discarding of less significant leads and the analysis of the leads with analysis of the most informative of them which can be useful for detecting cardiovascular diseases or optimize the study when performing an ECG. In such direction in this study the transfer entropy and the mutual information of the 12 leads of ECGs were calculated.

IV. Results

Having the ECG recordings, the mutual information between the 12 leads of both databases was calculated using the algorithm created by John M. O' Toole [20]. In Fig. 3 and Fig. 4, the results for twelve recordings randomly selected are shown. It should be noticed that the lack of orientation of mutual information is reflected in the symmetry of the matrices shown in both figures. These two cases reveal structures of relationship between leads with greater intensity both in the upper-left quadrant and in the lower-right quadrant. By analyzing the intensity of the color scale, it can be noted a greater link of information between determined leads, for example, II with aVR and aVF, aVR with V1 to V6 or V5 with V4 and V6. In order to simplify the analysis, the mean mutual information matrix was calculated for each database. The obtained results can be seen in Fig. 1. The corresponding mean mutual information matrices show similar structures to the ones found in the selected examples in Fig. 3 and Fig. 4. In all these figures, the maximum value is found in the diagonal of the matrix, which expresses the link between the information of a given lead and itself. Since this value can be twice or thrice bigger than the maximum value between two different leads, the color scale was limited in order to better highlight the results.

The results obtained by calculating the mean transfer entropy matrix for each databases are shown in Fig. 2 . The transfer entropy for each individual recording was calculated following the Darbellay-Vajda partitioning algorithm developed by Joon Lee [21]. The highest intensity of the information flow given by transfer entropy is found in a horizontal strip formed between leads V1 to V4, and a vertical strip formed between leads V1 to V3. The minimum value can be found in the matrix diagonal since there is no information flow between a lead and itself. The color scale used in these figures was normalized in order to compare the results between both databases, revealing a great coincidence between them.

It is not possible to appreciate in plain sight the obtained results for Fig. 5 and Fig. 6. However, a more detailed inspection over these examples reveal a relationship between lead V2 with leads V3 to V6 in most cases for both databases, showing a high transfer entropy value.

It should be noted in all figures that the designations aVR, aVF and aVL are equivalent to VR, VF and VL respectively.

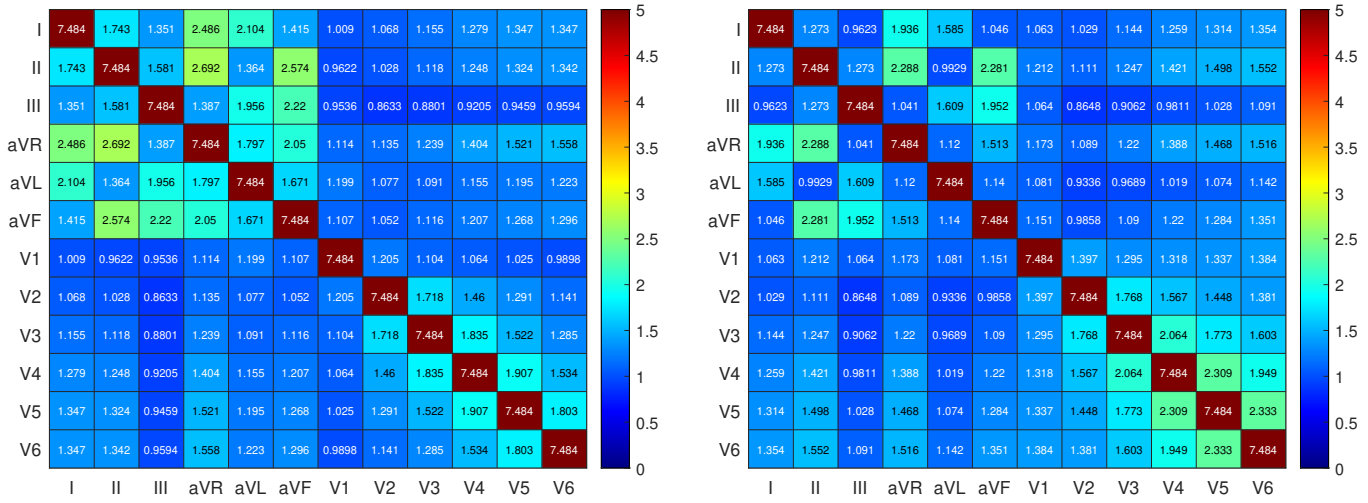


Fig. 1. Mutual information average of each database

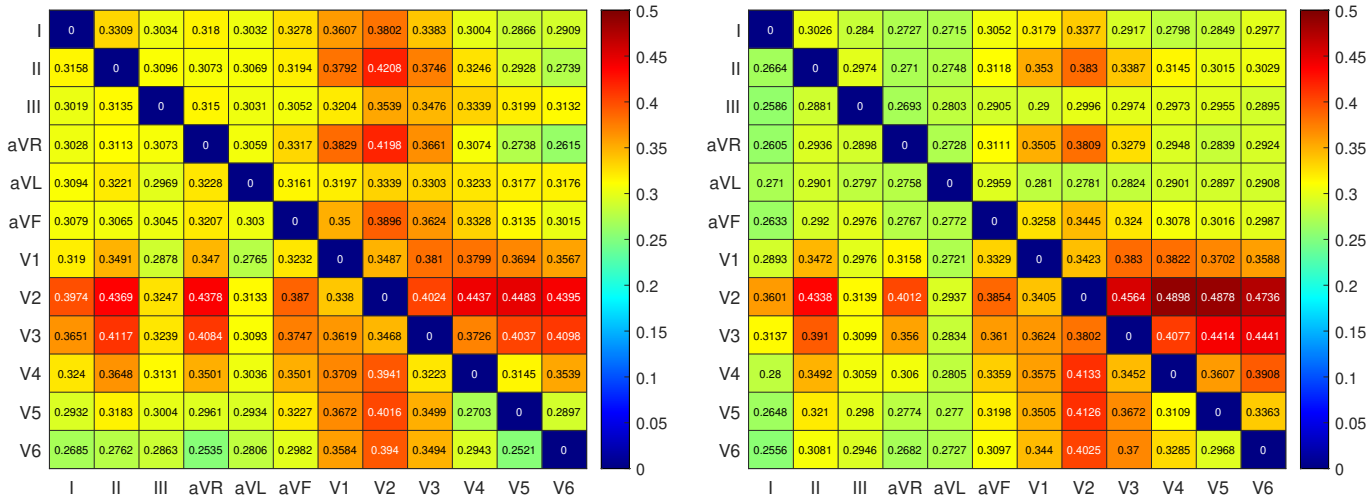


Fig. 2. Transfer Entropy average of each database

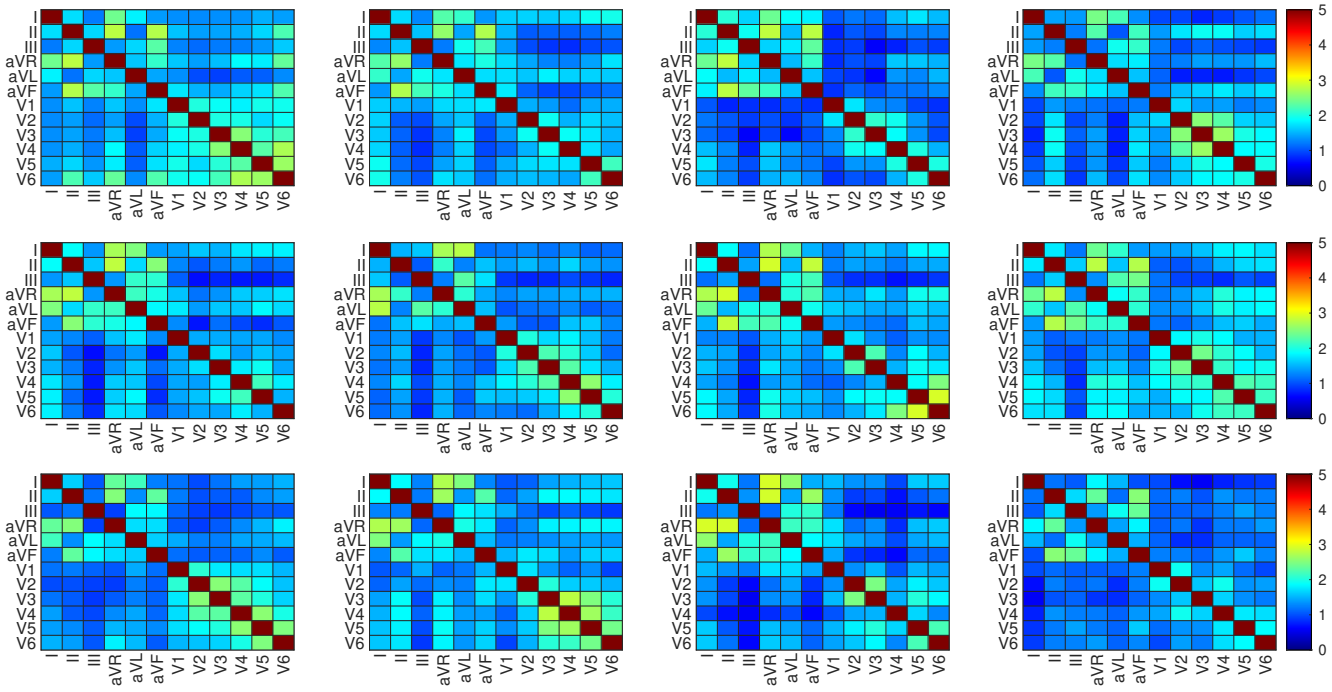


Fig. 3. Mutual Information on 12 randomly selected ECG recordings from the ICBEB 2018 database

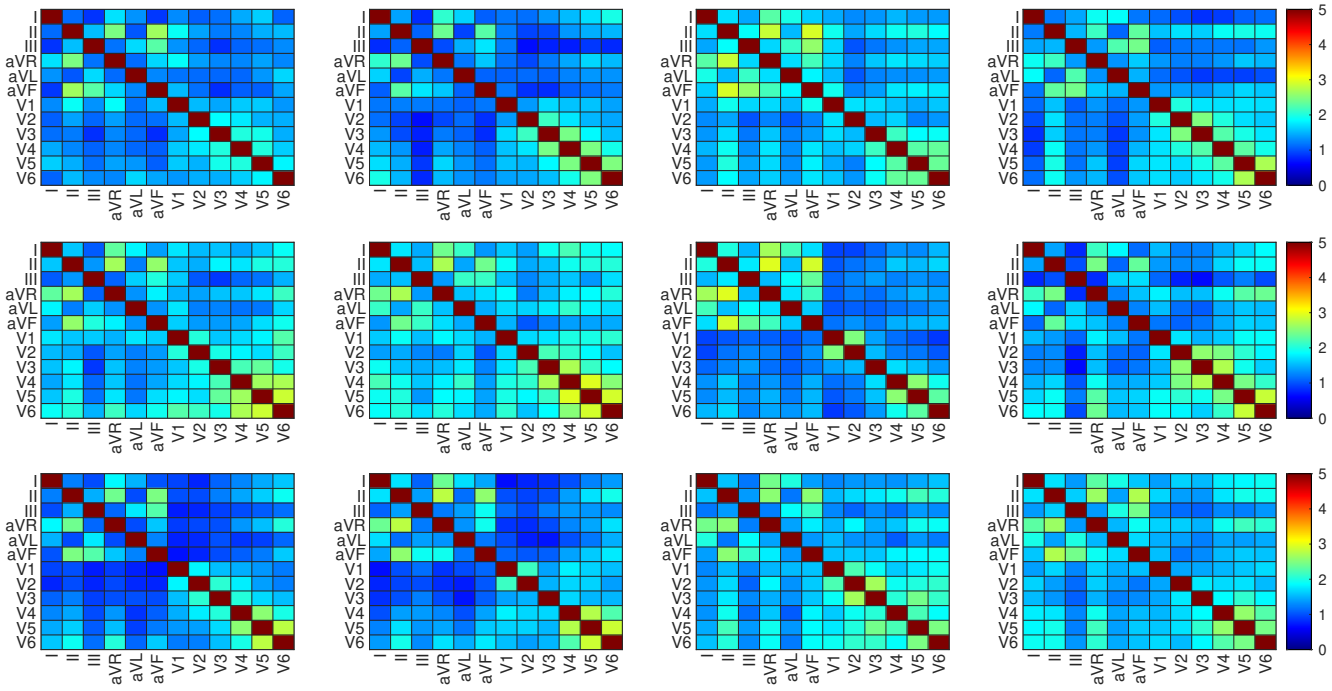


Fig. 4. Mutual Information on 12 randomly selected ECG recordings from the PTB-XL database

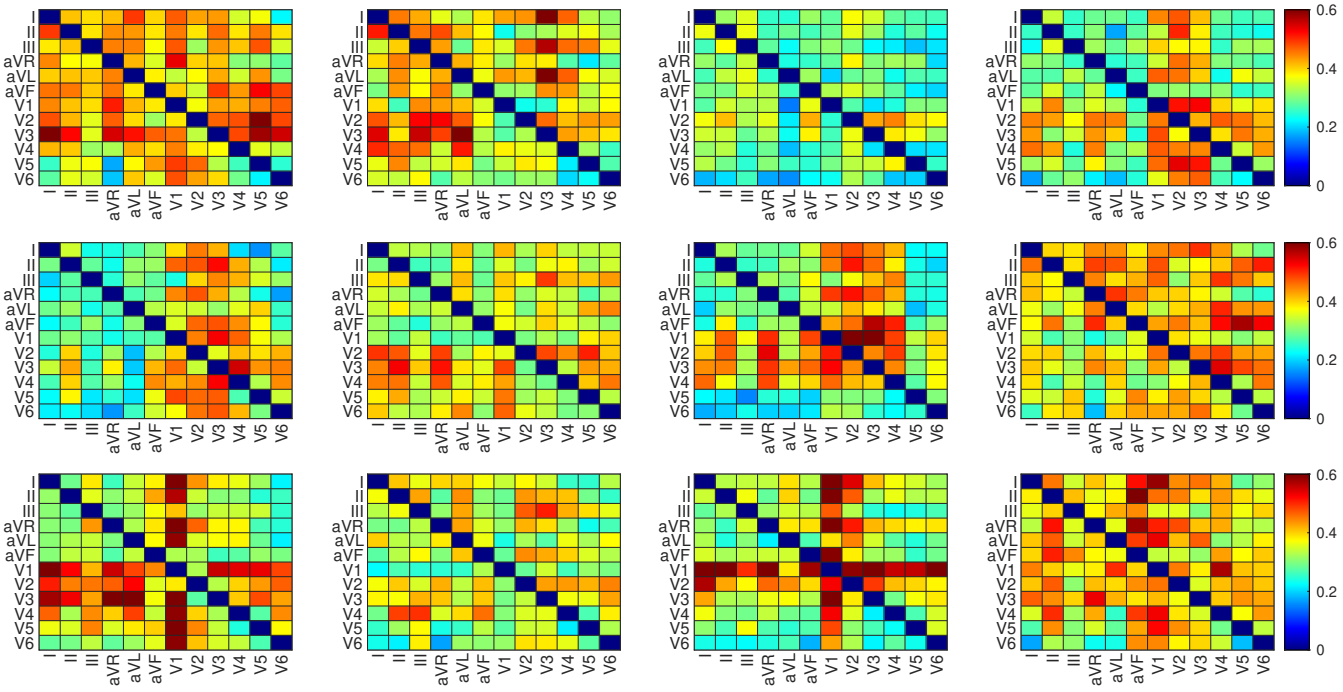


Fig. 5. Transfer Entropy on 12 randomly selected ECG recordings from the ICBEB 2018 database

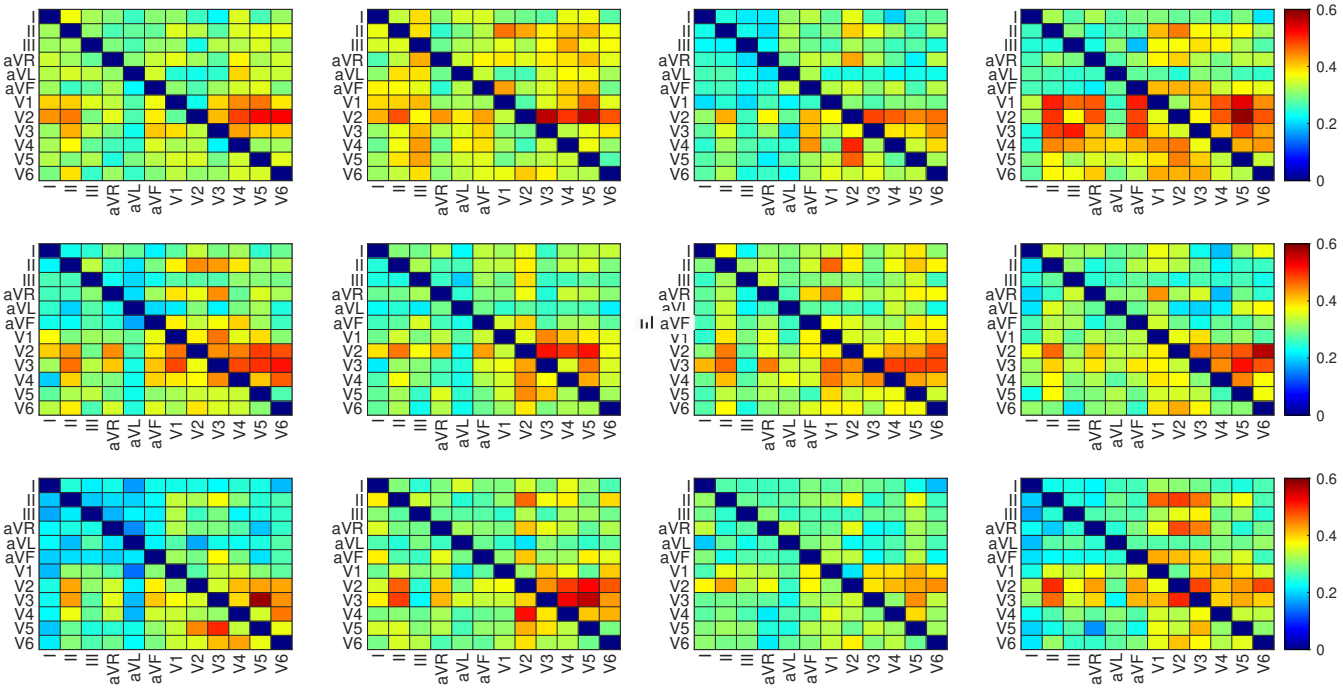


Fig. 6. Transfer Entropy on 12 randomly selected ECG recordings from the PTB-XL database

V. Conclusions

The objective of the present investigation has been fulfilled in showing in a methodical way how the leads of an ECG are linked by means of information theory tools applied to signal analysis of direct ECG records. By using two databases of different origin, the results obtained have a validation that aims to continue with the line of research. The mutual information matrix shows that it is possible to identify pairs of ECG leads with high dependencies, which could provide insights into the underlying physiological processes or help in reducing dimensionality for further analysis by removing or combining redundant leads that provide similar information. The information flux in the horizontal leads (V1-V6) shows in many cases a deep relation between them respect to the frontal leads, despite the frontal leads also being dependent between themselves. Given that the leads in any particular recording are not perfectly in sync, transfer entropy is a measure that does not reveal much relevant information by looking at each recording individually. However, it does prove that there is a clear information flow between certain leads by looking at the average resulting transfer entropy matrix for each database.

In reference to mutual information the highest intensity of information is given among the leads I to aVF constituting an square in the top left corner of the mean mutual information matrix, while the other highest values are located in a square in the bottom right corner of the same matrix limited by the V2 lead. In the case of transfer entropy, the highest values can be found in the intersections of the lead V2 with leads I, II, aVR, aVF, V3-V6, constituting a cross of a more intense color near the center of the matrix. It can be concluded from this study that the best leads to be used in a two-lead ECG analysis would be leads II/III and lead V2. This could be useful in studies where there is a need to reduce the amount of leads used. In the literature there are examples of researches developed using a single lead, as the case of Plawiak [22] who classifies ECG signals according to their respective pathology and Mathews et al. who has worked in deep learning with ECG single lead signals [23].

In order to improve these results the data could be preprocessed in order to synchronise and normalise all the data used, and in such a way, the results obtained could be enhanced. Although this implies that the signal processing will require more steps, the mutual information and transfer entropy provide great efficiency in computational implementation and performance, therefore it is a sensible first stage to improve the results.

References

[1] A. Bayés De Luna, "Basic Electrocardiography: Normal and Abnormal ECG Patterns", John Wiley, 2008.

[2] R.K. Tripathy, A. Bhattacharyya and R.B. Pachori, "A novel approach for detection of myocardial infarction from ECG signals of multiple electrodes", IEEE Sensors Journal, IEEE, vol.19, no.12, pp.4509–4517, 2019.

[3] L.Xie, Z.Li, Y.Zhou, Y.He and J.Zhu, "Computational diagnostic techniques for electrocardiogram signal analysis", Sensors, MDPI, vol.20, no.21, p. 6318, 2020.

[4] S.K. Berkaya, A.K. Uysal, E.S. Gunal, S. Ergin, S. Gunal and M.B. Gulmezoglu, "A survey on ECG analysis", Biomedical Signal Processing and Control, Elsevier, vol.43, pp.216–235, 2018.

[5] R. Sameni, F. Vrins, F. Parmentier, C. Herail, V. Vigneron, M. Verleysen, C. Jutten and M.B. Shamsollahi, "Electrode selection for noninvasive fetal electrocardiogram extraction using mutual information criteria", arXiv preprint arXiv:2302.00206, 2023.

[6] J. Runge, S. Bathiany, E. Bollt et al., "Inferring causation from time series in Earth system sciences", Nature communications, vol.10, no. 1, p.2553, 2019. [Online]. Available: https://doi: 10.1038/s41467-019-10105-3.

[7] T. Craciunescu, A. Murari and M. Gelfusa, "Causality Detection Methods Applied to the Investigation of Malaria Epidemics", Entropy, issn:1099-4300, vol.21, no.8, 2019. [Online]. Available: https://doi: 10.3390/e21080784

[8] A. C. Yang, C.-K. Peng and N. E. Huang, "Reply To: Comments on identifying causal relationships in nonlinear dynamical systems via empirical mode decomposition", Nature communications, vol.13, no.1, pp.1–3, 2022. [Online]. Available: https://doi:10.1038/s41467-022-30360-1.

[9] A. Kraskov, H. Stögbauer, and P. Grassberger, "Estimating mutual information", "PHYSICAL REVIEW E", vol. 69(6), pp.66138-1–66138-16, 2004.

[10] T.M. Cover and J.A. Thomas, "Elements of Information Theory", Wiley, 1991.

[11] A. Renyi, "Probability Theory", North Holland, Amsterdam, 1971.

[12] A.E. BozorgMagham, S. Motesharrei, S.G. Penny and E. Kalnay ,"Causality analysis: identifying the leading element in a coupled dynamical system", PLoS One, Public Library of Science San Francisco, CA USA, vol.10, p.e0131226, 2015.

[13] T. Bossomaier, L. Barnett, M. Harre, J.T. Lizier ,"Transfer entropy", Springer, 2016.

[14] D.Gençağa, "Transfer entropy", Entropy, MDPI, vol.20, no.4, p.228, 2018.

[15] J. Lee, S. Nemati, I. Silva, "Transfer Entropy Estimation and Directional Coupling Change Detection in Biomedical Time Series", BioMed Eng OnLine 11-19, 2012. [Online]. Available: https://doi.org/10.1186/1475-925X-11-19

[16] F.F. Liu, C.Y. Liu, L.N. Zhao, X.Y. Zhang, X.L. Wu, X.Y. Xu, Y.L. Liu, C.Y. Ma, S.S. Wei, Z.Q. He, J.Q. Li and N.Y. Kwee, "An open access database for evaluating the algorithms of ECG rhythm and morphology abnormal detection.", Journal of Medical Imaging and Health Informatics, vol.8(7), pp.1368–1373, 2018. [Online]. Available: http://2018.icbeb.org/Challenge.html

[17] P. Wagner, N. Strodthoff, R. Bousseljot, W. Samek and T. Schaeffter, "PTB-XL, a large publicly available electrocardiography dataset (version 1.0.2)", PhysioNet, 2022. [Online]. Available: https://doi.org/10.13026/zx4k-te85.

[18] P. Wagner, N. Strodthoff, R. Bousseljot, D. Kreiseler, F. Lunze, W. Samek, T. Schaeffter, "PTB-XL: A Large Publicly Available ECG Dataset", Scientific Data, 2020. [Online]. Available: https://doi.org/10.1038/s41597-020-0495-6

[19] A. Goldberger, L. Amaral, L. Glass, J. Hausdorff, P. Ivanov, R. Mark and H. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals", vol.101 (23), pp. e215–e220, 2000.

[20] J.M. O'Toole (2020) repository on GitHub. [Online]. Available: https://github.com/otoolej/mutual_info_kNN

[21] J. Lee, S. Nemati, I. Silva, "Transfer Entropy With Partitioning webpage on PhysioNet", 2016. [Online]. Available: https://physionet.org/content/tewp/1.0.0/

[22] P.Plawiak, "Novel methodology of cardiac health recognition based on ECG signals and evolutionary-neural system", Expert Systems with Applications, Elsevier, vol.92, pp 334–349, 2018.

[23] S.M. Mathews, C. Kambhamettu and K.E. Barner, "A novel application of deep learning for single-lead ECG classification, Computers in biology and medicine", Elsevier, vol.99, pp 53–62, 2018.